

MH2500 Probability and Introduction to Statistics

Slides for Week 9-2

Topics: Order Statistics

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Sum of Normal Variables

If $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim N(\mu_2, \sigma_2^2)$, then $X + Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.

Proof: WLOG, we can assume that $\mu_1 = \mu_2 = 0$. The density functions of X and Y are

$$\frac{1}{\sqrt{2\pi}\sigma_1} \exp\left(-\frac{x^2}{2\sigma_1^2}\right) \quad \text{and} \quad \frac{1}{\sqrt{2\pi}\sigma_2} \exp\left(-\frac{x^2}{2\sigma_2^2}\right).$$

The convolution is

$$f_Z(z) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma_1} \frac{1}{\sqrt{2\pi}\sigma_2} \exp\left(-\frac{x^2}{2\sigma_1^2} - \frac{(z-x)^2}{2\sigma_2^2}\right) dx$$

Let $h^2 = \sigma_1^2 + \sigma_2^2$:

$$\begin{aligned} f_Z(z) &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma_1} \frac{1}{\sqrt{2\pi}\sigma_2} \exp\left(-\frac{x^2}{2\sigma_1^2} - \frac{(z-x)^2}{2\sigma_2^2}\right) dx \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma_1} \frac{1}{\sqrt{2\pi}\sigma_2} \exp\left(-\frac{x^2(\sigma_1^2 + \sigma_2^2) - 2\sigma_1^2 zx + \sigma_1^2 z^2}{2\sigma_1^2 \sigma_2^2}\right) dx \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}h} \frac{1}{\sqrt{2\pi} \frac{\sigma_1 \sigma_2}{h}} \exp\left(-\frac{x^2 - 2\frac{\sigma_1^2}{h^2}zx + \frac{\sigma_1^2}{h^2}z^2}{2\frac{\sigma_1^2 \sigma_2^2}{h^2}}\right) dx \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}h} \frac{1}{\sqrt{2\pi} \frac{\sigma_1 \sigma_2}{h}} \exp\left(-\frac{(x - \frac{\sigma_1^2}{h^2}z)^2 - \frac{\sigma_1^4}{h^4}z^2 + \frac{\sigma_1^2}{h^2}z^2}{2\frac{\sigma_1^2 \sigma_2^2}{h^2}}\right) dx \\ &= \frac{1}{\sqrt{2\pi}h} \exp\left(-\frac{-\frac{\sigma_1^4}{h^4}z^2 + \frac{\sigma_1^2}{h^2}z^2}{2\frac{\sigma_1^2 \sigma_2^2}{h^2}}\right) = \frac{1}{\sqrt{2\pi}h} \exp\left(-\frac{z^2}{2h^2}\right) \end{aligned}$$

Sum of Normal Variables

Proposition

Let $X_i \sim N(\mu_i, \sigma_i^2)$ be independent normal variables. Then

$$\sum_i X_i \sim N\left(\sum_i \mu_i, \sum_i \sigma_i^2\right).$$

Conditional Distributions - Continuous Case

Definition

If X and Y are jointly continuous random variables, the **conditional density** of X given Y is defined to be

$$f_{X|Y}(x|y) = \begin{cases} \frac{f_{XY}(x, y)}{f_Y(y)}, & \text{if } f_Y(y) > 0; \\ 0, & \text{otherwise.} \end{cases}$$

Example

The **joint density** of X and Y is given by

$$f(x, y) = \begin{cases} \frac{12}{5}x(2 - x - y), & 0 < x < 1, 0 < y < 1; \\ 0, & \text{otherwise.} \end{cases}$$

Find the conditional density of X , given $Y = y$, where $0 < y < 1$.

Solution: For $0 < x < 1$,

$$\begin{aligned} f_{X|Y}(x|y) &= \frac{f(x, y)}{f_Y(y)} = \frac{f(x, y)}{\int_{-\infty}^{\infty} f(x, y) dx} = \frac{\frac{12}{5}x(2 - x - y)}{\int_0^1 \left(\frac{12}{5}x(2 - x - y) \right) dx} \\ &= \frac{x(2 - x - y)}{\frac{2}{3} - \frac{y}{2}} \\ &= \frac{6x(2 - x - y)}{4 - 3y}. \end{aligned}$$

Example

The joint density of X and Y is given by

$$f(x, y) = \begin{cases} e^{-x/y} e^{-y} / y, & \text{if } 0 < x, y < \infty; \\ 0, & \text{otherwise.} \end{cases}$$

Find $P\{X > 1 | Y = y\}$ (assuming that $y > 0$).

Solution:

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} = \frac{e^{-x/y} e^{-y} / y}{\int_0^{\infty} e^{-x/y} e^{-y} / y dx} = \frac{1}{y} e^{-x/y}$$

$$P\{X > 1 | Y = y\} = \int_1^{\infty} \frac{1}{y} e^{-x/y} dx = \left(-e^{-x/y} \right)_1^{\infty} = e^{-1/y}.$$

Order Statistics

Suppose $X_1, X_2, \dots, X_n$ are independent random variables with a common cdf F and density f .

- Let U denote the **maximum** of the X_i 's.

Then the cdf of U is given by

$$F_U(u) = P(U \leq u) = P(X_1 \leq u)P(X_2 \leq u) \cdots P(X_n \leq u) = (F(u))^n.$$

and so differentiating gives the density

$$f_U(u) = nf(u)(F(u))^{n-1}.$$

Likewise, let V denote the **minimum** of the X_i 's. Then

$$\begin{aligned}1 - F_V(v) &= P(V \geq v) \\&= P(X_1 \geq v)P(X_2 \geq v) \cdots P(X_n \geq v) = (1 - F(v))^n.\end{aligned}$$

Hence

$$F_V(v) = 1 - (1 - F(v))^n$$

and

$$\begin{aligned}f_V(v) &= (1 - (1 - F(v))^n)' = -n(1 - F(v))^{n-1}(-f(v)) \\&= nf(v)[1 - F(v)]^{n-1}.\end{aligned}$$

Example

Suppose that $X_1, X_2, \dots, X_n \sim \text{Exp}(\lambda)$ are independent. Find the distribution of $\min\{X_1, X_2, \dots, X_n\}$.

By the previous slide, the density is given by

$$f_V(v) = n(\lambda e^{-\lambda v})(e^{-\lambda v})^{n-1} = n\lambda \left(e^{-\lambda v} (e^{-\lambda v})^{n-1} \right) = (n\lambda) e^{-(n\lambda)v}.$$

Thus $V \sim \text{Exp}(n\lambda)$.

Joint distribution of Min and Max

Recall that $U = \max X_i$ and $V = \min X_i$. Consider the joint CDF $P\{U \leq u, V \leq v\}$.

- When $u \leq v$,

$$P\{U \leq u, V \leq v\} = P\{U \leq u\} = (F(u))^n$$

- When $u \geq v$,

$$\begin{aligned} P\{U \leq u, V \leq v\} &= P\{U \leq u\} - P\{U \leq u, V \geq v\} \\ &= (F(u))^n - (F(u) - F(v))^n \end{aligned}$$

Hence the joint density function is

$$f(u, v) = \begin{cases} 0, & u \leq v; \\ n(n-1)f(u)f(v)(F(u) - F(v))^{n-2}, & u \geq v. \end{cases}$$

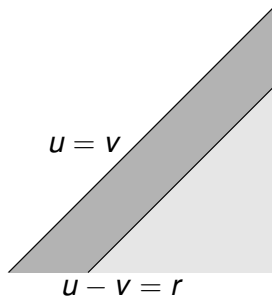
- If the X_i 's are uniform on $[0, 1]$, then

$$f(u, v) = n(n-1)(u-v)^{n-2}, \quad 1 \geq u \geq v \geq 0.$$

Question

Let $R = U - V$ be the range of $X_1, \dots, X_n$. What is the density function of the range for uniform random variables?

Solution: Find the CDF of $R = U - V$ first and then differentiate it.



$$\begin{aligned}
 P\{R \leq r\} &= P\{U - V \leq r\} \\
 &= \int_{-\infty}^{\infty} \int_v^{v+r} f(u, v) du dv \\
 &= \int_{-\infty}^{\infty} \int_0^r f(v + s, v) ds dv \\
 &= \int_0^r \int_{-\infty}^{\infty} f(v + s, v) dv ds
 \end{aligned}$$

$$\frac{d}{dr} P\{R \leq r\} = \frac{d}{dr} \int_0^r \int_{-\infty}^{\infty} f(v + s, v) dv ds$$

which gives

$$f_R(r) = \int_{-\infty}^{\infty} f(v + r, v) dv.$$

Joint density of (max, min) proved just now:

$$f(u, v) = n(n-1)(u-v)^{n-2}, \quad 1 \geq u \geq v \geq 0.$$

Then

$$f_R(r) = \int_0^{1-r} f(v+r, v) dv = n(n-1)r^{n-2}(1-r), \quad 0 \leq r \leq 1.$$